

Experiment 6

AIM: Set Up NGINX Load Balancing

```
C:\Users\kisha>mkdir load-balance-demo
```

```
C:\Users\kisha>cd load-balance-demo
```

```
C:\Users\kisha\load-balance-demo>npm init -y
```

Wrote to C:\Users\kisha\load-balance-demo\package.json:

```
{
  "name": "load-balance-demo",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "description": ""
}
```

```
C:\Users\kisha\load-balance-demo>npm install express
```

added 65 packages, and audited 66 packages in 4s

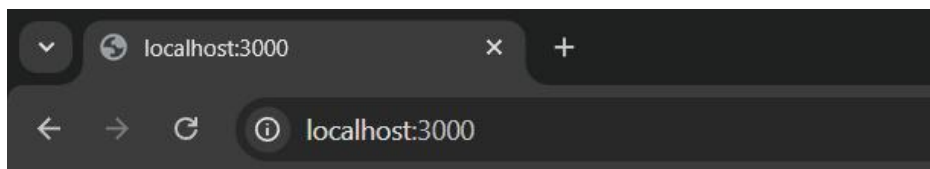
22 packages are looking for funding
run `npm fund` for details

found 0 vulnerabilities

```
C:\Users\kisha\load-balance-demo>notepad server.js
```

```
C:\Users\kisha\load-balance-demo>node server.js
```

Server running on port 3000



Response from Server running on port 3000

```
C:\Users\kisha\load-balance-demo>cd load-balancing
```

```
C:\Users\kisha\load-balance-demo\load-balancing>cd server1
```

```
C:\Users\kisha\load-balance-demo\load-balancing\server1>npm init -y
```

Wrote to C:\Users\kisha\load-balance-demo\load-balancing\server1\package.json:

```
{
  "name": "server1",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1",
    "start": "node server.js"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "dependencies": {
    "express": "^5.2.1"
  },
  "devDependencies": {},
  "description": ""
}
```

```
C:\Users\kisha\load-balance-demo\load-balancing\server1>npm install express
```

up to date, audited 66 packages in 1s

22 packages are looking for funding

run `npm fund` for details

found 0 vulnerabilities

```
C:\Users\kisha\load-balance-demo\load-balancing\server1>notepad server.js
```

```
const express = require('express');
const app = express();
```

```
const PORT = process.env.PORT || 3000;
```

```
// Route
```

```
app.get('/', (req, res) => {
  res.send('Response from Server 1 (Port 3000)');
});
```

```
// Start server
```

```
app.listen(3000, () => {  
  console.log('Server 1 running on port 3000');  
});
```

```
C:\Users\kisha\load-balance-demo\load-balancing\server1>cd ..
```

```
C:\Users\kisha\load-balance-demo\load-balancing>cd server2
```

```
C:\Users\kisha\load-balance-demo\load-balancing\server2>npm init -y
```

Wrote to C:\Users\kisha\load-balance-demo\load-balancing\server2\package.json:

```
{  
  "name": "server2",  
  "version": "1.0.0",  
  "main": "index.js",  
  "scripts": {  
    "test": "echo \"Error: no test specified\" && exit 1"  
  },  
  "keywords": [],  
  "author": "",  
  "license": "ISC",  
  "dependencies": {  
    "express": "^5.2.1"  
  },  
  "devDependencies": {},  
  "description": ""  
}
```

```
C:\Users\kisha\load-balance-demo\load-balancing\server2>npm install express
```

up to date, audited 66 packages in 1s

22 packages are looking for funding

run `npm fund` for details

found 0 vulnerabilities

```
C:\Users\kisha\load-balance-demo\load-balancing\server2>notepad server.js
```

```
const express = require('express');  
const app = express();
```

```
const PORT = process.env.PORT || 3001;
```

```
// Route
```

```
app.get('/', (req, res) => {
```

```
res.send('Response from Server 2 (Port 3001)');
});
```

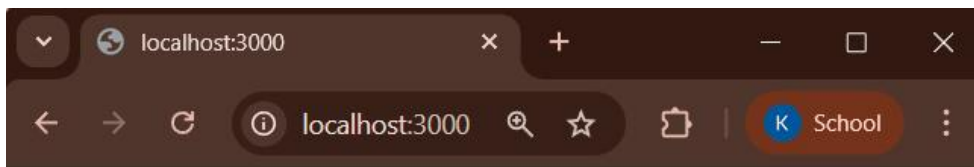
```
// Start server
app.listen(3001, () => {
  console.log('Server 2 running on port 3001');
});
```

```
C:\Users\kisha\load-balance-demo\load-balancing\server2>node server.js
Server 2 running on port 3001
```

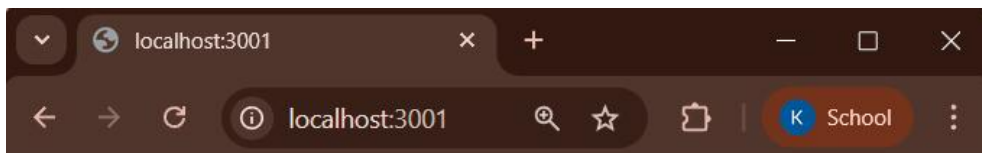
```
C:\Users\kisha\load-balance-demo\load-balancing\server2>cd ..
```

```
C:\Users\kisha\load-balance-demo\load-balancing>cd server1
```

```
C:\Users\kisha\load-balance-demo\load-balancing\server1>node server.js
Server 1 running on port 3000
```



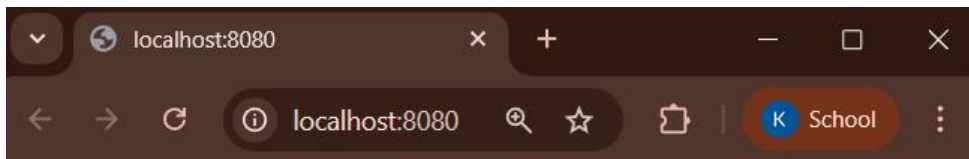
Response from Server 1 (Port 3000)



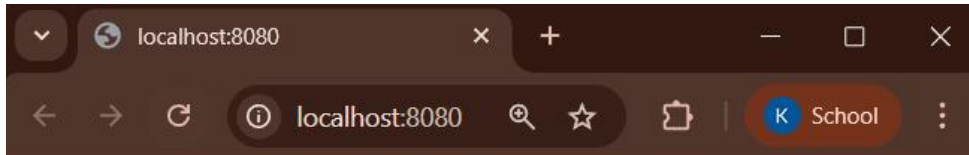
Response from Server 2 (Port 3001)

```
C:\Users\kisha\load-balance-demo>cd ..
C:\Users\kisha>cd..
C:\Users>cd..
C:\>cd nginx-1.29.5
C:\nginx-1.29.5>cd conf
C:\nginx-1.29.5\conf>nginx.conf
worker_processes 1;
events {
  worker_connections 1024;
}
http {
  upstream myapp {
```

```
server localhost:3000;
server localhost:3001;
}
server {
    listen 8080;
    server_name localhost;
    location / {
        proxy_pass http://myapp;
    }
}
}
C:\nginx-1.29.5\conf>cd ..
C:\nginx-1.29.5>nginx -s stop
C:\nginx-1.29.5>nginx
```



Response from Server 1 (Port 3000)



Response from Server 2 (Port 3001)

```
C:\Users\kisha\load-balance-demo\load-balancing\server1> notepad server.js
const express = require("express");
const app = express();
const PORT = 3000;
app.get("/", (req,res)=>{
    setTimeout(()=>{
        res.send("Slow Server - Port 3000");
    },5000);
});
app.listen(PORT,()=>{
    console.log("Server 1 running on port 3000");
});
C:\Users\kisha\load-balance-demo\load-balancing\server1> node server.js
Server 1 running on port 3000
```

C:\Users\kisha\load-balance-demo\load-balancing\server2> notepad server.js

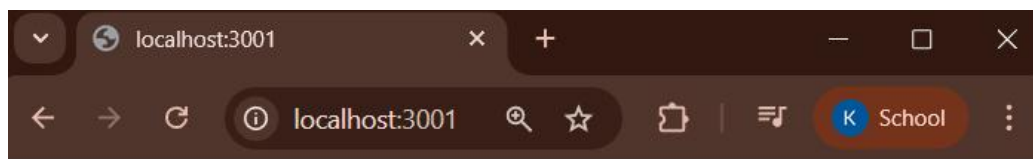
```
const express = require("express");
const app = express();
const PORT = 3001;
app.get("/",(req,res)=>{
res.send("Fast Server - Port 3001");
});
```

```
app.listen(PORT,()=>{
console.log("Server 2 running on port 3001");
});
```

C:\Users\kisha\load-balance-demo\load-balancing\server2> node server.js
Server 2 running on port 3001



Slow Server - Port 3000



Fast Server - Port 3001

C:\>cd nginx-1.29.5

C:\nginx-1.29.5>cd conf

C:\nginx-1.29.5\conf>nginx.conf

```
worker_processes 1;
```

```
events {
```

```
    worker_connections 1024;
```

```
}
```

```
http {
```

```
    upstream myapp {
```

```
        server localhost:3000;
```

```
        server localhost:3001;
```

```
    }
```

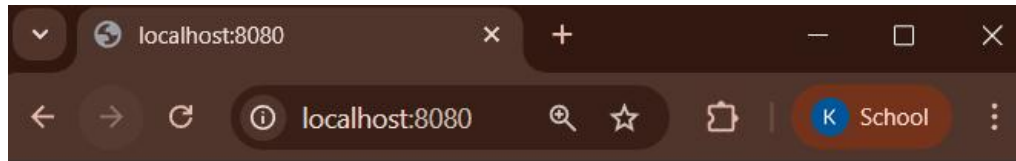
```
server {
```

```
    listen 8080;
```

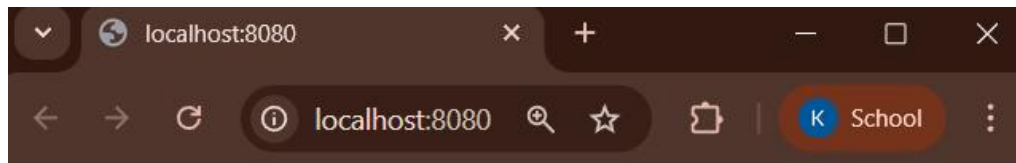
```
    server_name localhost;
```

```
    location / {
```

```
    proxy_pass http://myapp;
  }
}
C:\nginx-1.29.5\conf>cd ..
C:\nginx-1.29.5>nginx
```



Slow Server - Port 3000



Fast Server - Port 3001